Solutions
QA : Geometry Review Test

CEX-Q-0240/25

1	1	**2**	4	**3**	4	**4**	1	**5**	4	**6**	2	**7**	2	**8**	3	**9**	4	**10**	2
11	4	**12**	3	**13**	1	**14**	3	**15**	2	**16**	4	**17**	–	**18**	2	**19**	1	**20**	1
21	3	**22**	4	**23**	2	**24**	3	**25**	1										

1. 1 Radius of the sphere is r
From the centre of the sphere, radius of the upper part $= \sqrt{r^2 - h^2}$

Hence, $4\pi r^2 + 2\pi(r^2 - h^2) = \frac{125}{100}4\pi r^2$

$\Rightarrow 6r^2 - 2h^2 = 5r^2 \Rightarrow h = \frac{r}{\sqrt{2}}$

2. 4

Rope = 150 ft
Half of rope = 75 ft
Pole = 100 ft
Distance above the ground = 25 ft
Here, half of rope = pole – distance above the ground
Nothing left to make the curve

3. 4 Side of square = 25
Area of rhombus = 500

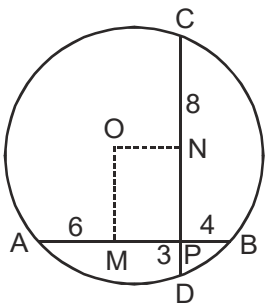
So, altitude, MD $= \frac{500}{25} = 20$

$MF = \sqrt{25^2 - 20^2} = 15$

Area $(\Delta MFD) = \frac{1}{2} \times 15 \times 20 = 150$

$\therefore$ area of shaded region = 625 – (500 – 150) = 275

4. 1



$6 \times 4 = CP \times 3 \Rightarrow CP = 8$
MP = 1
ON = 1
CD = 11
CN = 5.5
$ON^2 + CN^2 = OC^2$ (OC = r)
$\Rightarrow r^2 = 1^2 + (5.5)^2$
$\Rightarrow r^2 = 31.25$

So area $= \pi r^2$

$= \frac{125}{4}\pi$

5. 4 As, tangents from an exterior point make right angle with the radius $\Rightarrow$ AOBX & ODX'E are squares.
$\therefore$ AO = BO = 40 cm and OD = DE = 30 cm

Length of string wound around the circle $X = \frac{3}{4} \times 80\pi$

$= 60\pi$

Similarly, wound around circle $X' = \frac{3}{4} \times 60\pi$

So, total length $= 40 + 40 + 30 + 30 + 60\pi + 45\pi$

$\Rightarrow 140 + 105\pi.$

6. 2 Let radius of the circles with centres C_1 and C_2 = R and radius of the circle at centre C_1 = r.

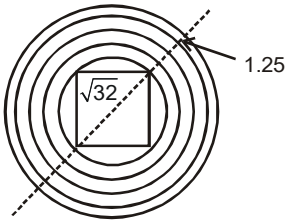
$\because \frac{1}{\sqrt{r}} = \frac{1}{\sqrt{R}} + \frac{1}{\sqrt{R}}$

$\frac{1}{\sqrt{10}} = \frac{2}{\sqrt{R}} \Rightarrow R = 40$ cm

7. 2 The possible triangles are with sides 4, 4, 8 or 8, 8, 4 but triangle one with sides 4, 4, 8 is not possible.
So perimeter is 8 + 8 + 4 = 20

8. 3 Let area of backyard be x^2 this year and y^2 last year.
then $x^2 - y^2 = 131$
$\Rightarrow (x - y)(x + y) = 1 \times 131$
$\Rightarrow x = 66$ and $y = 65$
So, number of tomatoes produced this year is 66^2 = 4356

9. 4



Diagonal $= \sqrt{32} \times \sqrt{2} = 8$

Since, the circles are equally spaced at 1.25 cm, the distance between innermost and outermost circle = 5 cm

$\therefore$ diameter of outermost circle = 5 + 8 + 5 = 18 cm

or $a\sqrt{2} = 18 \Rightarrow a = 9\sqrt{2} \Rightarrow \text{area} = (9\sqrt{2})^2 = 162$

10. 2 ab = 4

$\Rightarrow$ ab = 2 × 2 or 4 × 1

221, 222, 223, 441

these are possible triangles

22x will be right angled if x is 1, 2 or 3

44x is a triangle only if x = 1

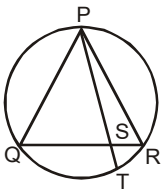
$\Rightarrow$ 221 is acute $1^2 + 2^2$, 222 is equilateral 223 is obtuse

144 is acute.

Only triangle 223 is possible

Hence c = 3

11. 4



As S is not the circumcentre,

$PS \neq ST$ and $QS \neq SR$

PS × ST = QS × SR … (i)

We know that Arithmetic mean $\geq$ Geometric mean

$$\Rightarrow \frac{PS + ST}{2} \geq \sqrt{PS \times ST}$$

Since $PS \neq ST$,

$$\frac{PS + ST}{2} > \sqrt{PS \times ST}$$

$$\Rightarrow \frac{PS + ST}{2} > \sqrt{QS \times SR}$$

$$\Rightarrow \frac{PS + ST}{PS \times ST} > \frac{2\sqrt{QS \times SR}}{QS \times SR}$$

$$\Rightarrow \frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{QS \times SR}} \quad \cdots \text{(ii)}$$

Again, $\frac{QS + SR}{2} > \sqrt{QS \times SR}$

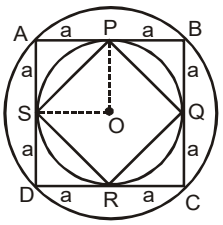
$$\Rightarrow \frac{2}{QR} < \frac{1}{\sqrt{QS \times SR}}$$

$$\Rightarrow \frac{4}{QR} < \frac{2}{\sqrt{QS \times SR}}$$

$$\Rightarrow \frac{1}{PS} + \frac{1}{ST} > \frac{2}{\sqrt{QS \times SR}} > \frac{4}{QR} \quad \cdots \text{From (ii)}$$

12. 3 Radius of the smaller circle must be less than 1 which is option (3).

13. 1



PQRS would also be a square

radius of big circle = diagonal of the square APOS

$= \sqrt{2}a$

Circumference $= 2\pi\sqrt{2}a$

Now, radius of small circle = side of square APOS = a

Consider ΔPSO, $PS^2 = OP^2 + OS^2 \Rightarrow PS = \sqrt{2}a$

Circumference of square with side $a\sqrt{2} = 4\sqrt{2}a$

$\Rightarrow \text{Ratio} = 2\pi a\sqrt{2} : 4a\sqrt{2} = \pi : 2$

14. 3 Join AY

L2et AX = a

Diameter = (a + 3)

Since AB is the diameter $\angle AYB = 90°$

$\Rightarrow \angle BXY = 90°$

$\Rightarrow BY = \sqrt{3^2 + 5^2} = \sqrt{34}$

Consider triangles ABY and BXY

$\angle ABY = \angle XBY$ and both triangles are right angled.

$\therefore$ They are similar Δs and sides are proportional

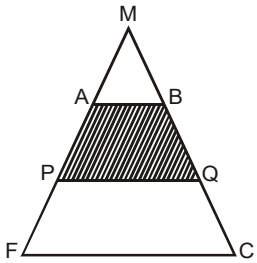
$$\frac{AB}{BY} = \frac{BY}{BX} = \frac{AY}{XY}$$

$$\frac{(a+3)}{\sqrt{34}} = \frac{\sqrt{34}}{3} \Rightarrow a = \frac{25}{3}$$

$$\text{Diameter} = \frac{25}{3} + 3 = \frac{34}{3}$$

$$\Rightarrow \text{Circumference} = \frac{34\pi}{3}\text{cm}$$

15. 2 Extending FA and CB to meet at M as shown below:



AB = 3a (side length of hexagon)
$\Rightarrow$ FC = 6a
from similarity PQ = 4a
and all triangles ΔAMB, ΔPMQ and ΔFMC are equilateral and similar to each other.
so, area of trapz. (AFCB) = ar(ΔMFC) – ar(ΔMAB)

$$= \frac{\sqrt{3}}{4}(6a)^2 - \frac{\sqrt{3}}{4}(3a)^2 = \frac{\sqrt{3}}{4} \times 27a^2$$

and shaded area = ar(ΔMPQ) – ar(ΔMAB)

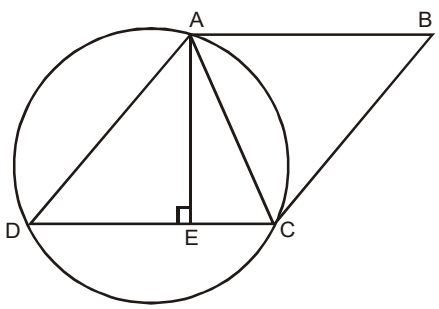
$$= \frac{\sqrt{3}}{4}(4a)^2 - \frac{\sqrt{3}}{4}(3a)^2 = \frac{\sqrt{3}}{4} \times 7a^2$$

So, required ratio $= \dfrac{(\sqrt{3}/4) \times 7a^2}{(\sqrt{3}/4) \times 27a^2} = \dfrac{7}{27}$

Alternatively:

Or, for similar Δs, required ratio $= \dfrac{4^2 - 3^2}{6^2 - 3^2} = \dfrac{7}{27}$

16. 4

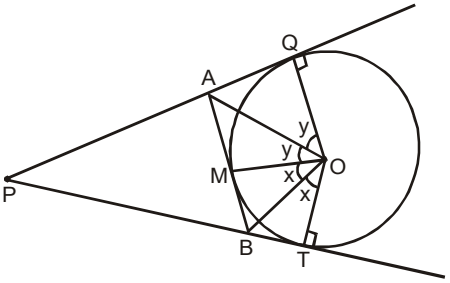


Area = 160 cm^2
DC × AE = 160
AE = 8 cm
and AD = 10 cm
$\Rightarrow$ DE = 6 cm {from Pythagorean triplets}
$\Rightarrow$ CE = 14 cm
Now, AC^2 = AE^2 + EC^2 = 8^2 + 14^2
$\Rightarrow$ AC = $\sqrt{260}$

17. Since, we know that here
Radius = R = $\perp$r distances from the origin to the tangent
Thus, $\perp$r distance from 0(0, 0) to the line
$2x - \sqrt{3}y + 4\sqrt{3} = 0$

$$\Rightarrow R = \left| \frac{2(0) - \sqrt{3}(0) + 4\sqrt{3}}{\sqrt{2^2 + (\sqrt{3})^2}} \right| = \frac{4\sqrt{3}}{\sqrt{7}}$$

18. 2



$\angle$QOT = 120°
Let, the tangent AB touches the circle at M
We join MO to get kites OMBT and OMAQ.
Now, we again join BO and AO such that
$\angle$TOB = $\angle$BOM = x and $\angle$MOA = $\angle$AOQ = y
and we know x + x + y + y = 120°

$\Rightarrow x + y = \dfrac{120°}{2} \Rightarrow \angle BOA = 60°$

19. 1 $\tan\alpha = 1 \Rightarrow \alpha = 45°$

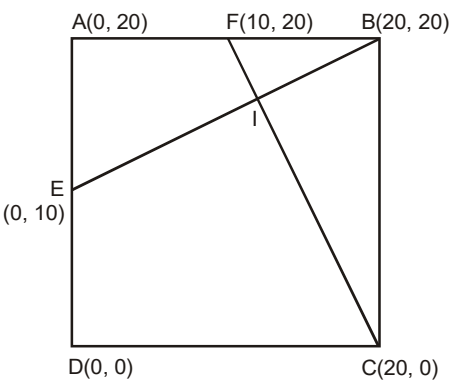
$\tan\beta = \dfrac{1}{2}$ and $\tan\gamma = \dfrac{1}{3}$

Now, we know α = 45° and we've to calculate ($\beta + \gamma$)

So, $\tan(\beta+\gamma) = \dfrac{\tan\beta + \tan\gamma}{1 - \tan\beta.\tan\gamma} = \dfrac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2}.\frac{1}{3}} = \dfrac{\frac{5}{6}}{\frac{5}{6}} = 1$

$\Rightarrow \beta + \gamma = 45°$
Then $\alpha + \beta + \gamma$ = 45° + 45° = 90°.

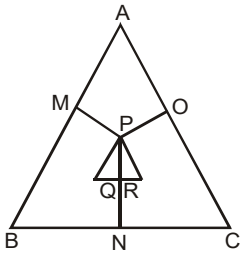
20. 1 On solving by co-ordinate geometry.



We consider the square is placed in a co-ordinate axes
So, equation of CF
2x + y = 40
and equation of BE
x = 2y – 20
On solving both the equations, we get, y-coordinate of point I would be y = 16.
$\Rightarrow$ Height of ΔFBI = 20 – 16 = 4 and Base = 10

So, area $= \dfrac{1}{2} \times 4 \times 10 = 20$ sq. unit

21. 3 Since, we know that the sum of perpendicular distances from any point inside the triangle to each side of the triangle is equals to the height of the equilateral triangle.
So, we move lines RO and QN to form a new figure as



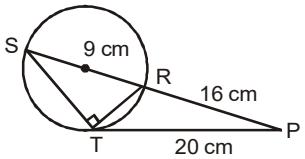
Now $PQ = \frac{\sqrt{3}}{2}$

So, MP + PO + (PQ + QN) $= \frac{\sqrt{3}}{2}(\sqrt{3}+1)$

$MP + PO + \frac{\sqrt{3}}{2} + QN = \frac{3}{2} + \frac{\sqrt{3}}{2}$

$\Rightarrow PM + QN + RO = \frac{3}{2}$

22. 4 $PR \times PS = PT^2$
$16 \times 25 = PT^2 \Rightarrow PT = 20$ cm
we join ST and RT
let, $\angle P = \alpha$
and $\angle RTP = \theta$
then $\angle RST = \theta$
$\Rightarrow \Delta PRT \sim \Delta PST$



$\frac{TR}{ST} = \frac{PR}{TP} \Rightarrow \frac{TR}{ST} = \frac{16}{20} = \frac{4}{5}$
$\Rightarrow 5.\ TR = 4.\ ST$
Now, from Pythagoras theorem
$ST^2 + TR^2 = 9^2$

$ST^2 + \left(\frac{4}{5} \times ST\right) = 9^2$

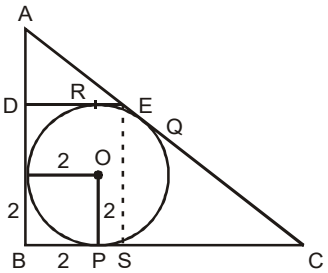
$\Rightarrow ST = \frac{45}{\sqrt{41}}$

Now, Area (ΔSTR) = $\frac{1}{2} \times TR \times ST$

$= \frac{1}{2} \times \frac{4}{5} \times ST^2 = \frac{4}{2 \times 5} \times \frac{45 \times 45}{41}$

Area (ΔSTR) = 20 cm^2

23. 2 Let the tangents BC and AC touch the circle at P and Q respectively.



So, from tangent property.
DR = 2 $\Rightarrow$ RE = 1 $\Rightarrow$ EQ = 1
let QC = x $\Rightarrow$ PC = x
Now, we draw a $\perp$r to BC from point E meeting at S
So, EC = (x + 1)
CS = (x – 1) and ES = 4
Now, from Pythagoras theorem $EC^2 = CS^2 + ES^2$
$(x + 1)^2 = (x - 1)^2 + 4^2 \Rightarrow x = 4$cm
So, BC = 6cm
$\because$ $\Delta ADE \sim \Delta ABC \Rightarrow AB = 8$cm
Thus, Area (ΔADE) = 1/2 × 3 × 4 = 6 cm^2.
Alternatively:
Since in radius = 2, which is possible for a right-angled triangle with perimeter 6,8,10
So, (BC = 6 and AB = 8) or (BC = 8 and AB = 6) 2nd pair satisfies the given options.

24. 3 From tangent property for a right-angled triangle, we know

r(in-radius) = $\frac{P + B - H}{2} = \frac{P + 16 - H}{2} = \frac{16 - (H - P)}{2}$

Again, $P^2 + B^2 = H^2$
$\Rightarrow$ $H^2 - P^2 = 16^2$
$\Rightarrow$ $(H - P)(H + P) = 16 \times 16$
So, possible values of (H – P) is {2,4,8}
$\Rightarrow$ possible in-radius would be {7, 6, 4}
Thus, only 2 possible even in-radius are possible

25. 1 We join QO such that ΔPQO becomes an equilateral triangle with area $25\sqrt{3}\,cm^2$

$\Rightarrow$ $\frac{\sqrt{3}}{4} \times (side)^2 = 25\sqrt{3}\,cm^2$

side = 10 cm {for ΔOPQ}
$\Rightarrow$ PQ = 10 cm
$\because$ PC : CQ = 4 : 1
$\Rightarrow$ PC = 8 and CQ = 2
Now, AC × BC = PC × CQ
$\Rightarrow$ AC × BC = 16
Now from AM $\geq$ GM

$\frac{AC + BC}{2} \geq (AC \times BC)^{1/2} \Rightarrow AC + BC \geq 8$

Thus, minimum possible length of AB = 8.